FAISAL AHMED

Junior Software Engineer | **Learning Shaper**



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LinkedIn | GitHub | Portfolio | learningshaper.com

"Junior Software Engineer at Learning Shaper (Learning Shaper Technologies Limited) and B.Sc. in Computer Science and Engineering Graduate from AIUB (CGPA: 3.08). Experienced in enterprise full-stack development (Java, Spring Boot, Laravel, React, C#, ASP.NET Core, Python, SQL) and Deep Learning / Computer Vision research (Hybrid ViT–GNN–TCN Frameworks for Sign Language & Facial Emotion AI)."

WORK EXPERIENCE

Learning Shaper (Learning Shaper Technologies Limited — learningshaper.com) — **Dhaka, Bangladesh**

Junior Software Engineer — **Oct 1, 2026 - Present**

- Developing and maintaining backend services, RESTful APIs, and web modules for **Learning Shaper**, an AI-powered School Management ERP & Learning Management System (LMS).
- Engineering core institutional workflows including automated grading, Question Banks, OMR scanning, attendance, and fee management with relational database integration.

Software Engineer Intern — **Jul 1, 2026 - Oct 1, 2026 · 3 months**

- Assisted in developing web interfaces, CRUD operations, and database integrations across academic and classroom management modules; promoted to full-time Junior Software Engineer.

TECHNICAL SKILLS

Programming & Web: C++, C#, Python, Java, R, HTML, CSS, JavaScript, PHP
Frameworks: .NET Framework, ASP.NET MVC, ASP.NET Core, Entity Framework Core, Spring Boot, Laravel, React, Tkinter
Deep Learning & AI Research: PyTorch (CUDA), Vision Transformer (ViT), Graph Neural Networks (GCN/GAT), Dilated TCN, MediaPipe 3D
Database & Architecture: SQL Server, MySQL, PostgreSQL | OOP, N-Tier Architecture, Multi-Tenant SaaS, RESTful APIs
Tools & Graphics: Visual Studio, VS Code, XAMPP, Postman, CodeBlocks, Git, OpenGL, Basic Graphics Design

UNDERGRADUATE THESIS & ACADEMIC RESEARCH (AIUB)

B.Sc. Thesis: Beyond Hand Gestures — Hybrid ViT–GNN–TCN for Joint Sign Language & Facial Emotion Recognition — **October 2026**

- Engineered a dual-branch PyTorch framework uniting **Vision Transformer (ViT)**, **Graph Neural Networks (GNN)**, **Dilated Causal TCN**, and **MediaPipe 3D** (478 facial + 21 hand landmarks) under Supervisor **Md. Hamid Uddin** (Lecturer, CS, AIUB).
- Achieved **96.74% Test Accuracy (96.77% Macro-F1; 97.21% Peak Val Acc)** across 47 dynamic sign classes and **100% Val / 80.95% Test Accuracy** using a lightweight **0.91M-parameter** Landmark GNN–TCN facial emotion branch.

Research Paper: A Hybrid ViT–GNN–TCN Framework for Static & Video-Based American Sign Language Recognition — **January 2026**

- Proposed a two-stage ViT–GNN–TCN architecture achieving **99.65% Test Accuracy** on static ASL alphabet classification (~87,000 RGB images, 29 classes) and **87.60% Test Accuracy** on dynamic WLASL word-level video recognition (T=16 frames).

CORE SOFTWARE ENGINEERING PROJECTS

Perfume Decant Management App - Python, Tkinter — **March 2026**

- Built a business management application for inventory, sales, and expense tracking
- Automated record-keeping using Python and designed GUI using Tkinter

Restaurant Order & Table Management System - ASP.NET Core — **January 2026**

- Built a web application using .NET Core (N-tier architecture)
- Implemented CRUD operations and database integration (SQL Server) focused on scalable and maintainable code structure

Recipe Sharing Platform - PHP, MySQL — **October 2025**

- Built a dynamic web application for sharing recipes
- Implemented CRUD functionality, database integration, and designed frontend using HTML, CSS, and JavaScript

Cargo Flow – From Warehouse to Sea - C++, OpenGL — **March 2025**

- Built a cargo flow simulation using OpenGL
- Implemented animation and object movement to visualize real-world logistics process

Library Management System - C#, .NET, SQL Server — **November 2024**

- Built a system to manage books and user records
- Implemented CRUD operations with SQL Server and developed using C# and .NET framework

EDUCATION

B.Sc. in Computer Science and Engineering (Graduated) — **September 2022 – October 2026**
American International University-Bangladesh (AIUB) | **CGPA: 3.08** | **Thesis:** Hybrid ViT–GNN–TCN Sign & Emotion AI

Higher Secondary Certificate (HSC), Science — Dr. Mahbubur Rahman Mollah College — **May 2021** | **GPA: 5.00**

Secondary School Certificate (SSC), Science — Shamsul Haque Khan School & College — **March 2019** | **GPA: 5.00**

SOFT SKILLS

Problem-Solving | Time Management | Adaptability | Communication | Team Work | Quick Learner